

FALCON's Response to RFQ LANDMARK GROUP for Supply of Conveyor Automation System – Option 2

July 1, 2023



www.falconautotech.com



Kind Attention –

Mr. Bhaskara Rao & Mr. Baiju Pillai

Landmark Group

Offer Ref: F23-00303_Op2; Date 01.07.2023

Subject – Techno-Commercial Offer for Conveyor Automation

Dear Team,

We are pleased to submit our Techno-Commercial Offer in response to your RFQ

As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops and calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Automation Technologies and references.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

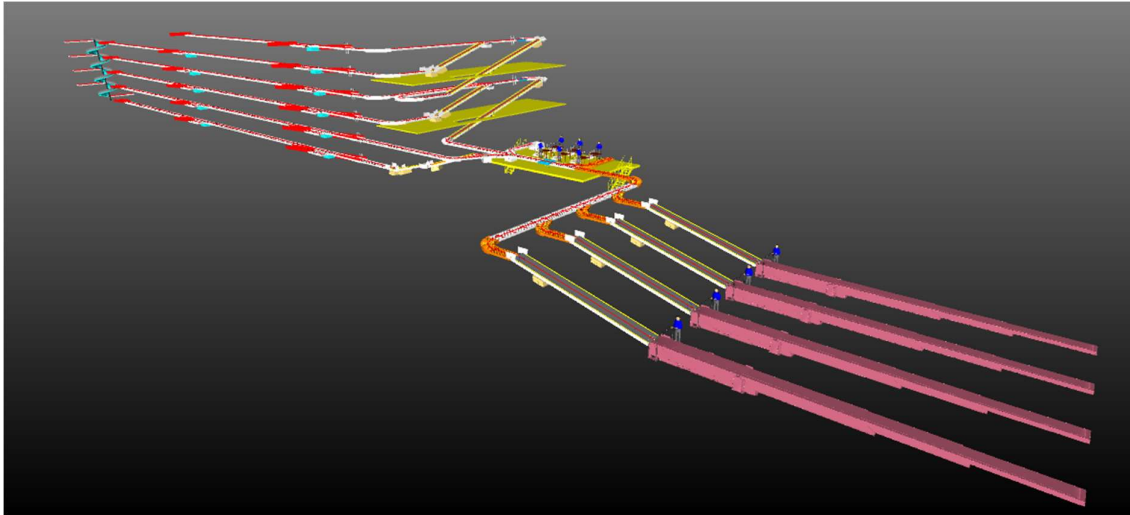
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Company Details

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Response to RFQ for Conveyor Automation_ Landmark Group, NCR



**Proposal Reference – F23-00303
Option-2**

**Falcon Autotech Private Limited
Plot No. 87, Sector Ecotech-1,
Extention-1, Greater Noida, Uttar
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1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Products Per Hour
3	DBO	Damaged Barcode
4	MEZZ	Mezzanine
5	AC	Alternating Current
6	DC	Direct Current
7	PLC	Programmable Logic Controller
8	IT	Information Technology
9	BOQ	Bill Of Quantity
10	I/O	Input/ Output
11	PDP	Power Distribution Panel
12	PC	Personal Computer
13	UPS	Uninterrupted Power Supply
14	SWEDI	Swivel Wheel Divert



2. About Falcon

THE COMPANY

ABOUT US

Falcon Autotech is one of the leading Intra-logistics automation solution providers with a sharp focus on Parcel, Piece and Case Handling applications. We have a broad portfolio of Sortation Systems for Retail, E-Commerce and CEP (Courier, Express and Postal) industries and have supplied our systems across various countries globally.

OUR VISION

To create a global impact on the society by facilitating world's leading organizations run their supply chains efficiently by leveraging technology enabled products and solutions and, in the process, establish Falcon Autotech as one of the world's most respected and trusted brands in Intra-logistics warehouse automation space.

OUR JOURNEY

The team started out in 2004 solving special purpose automation problems for clients and later pivoted to building a standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems. Over the last few years, the organization has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.

OUR CORE COMPETENCIES

Organizations are built and defined by the people they are able to attract, cultivate and retain and the talent management philosophy at Falcon Autotech could not agree more. At, Falcon Autotech we take a lot of pride in being the preferred career choice for some of the finest talent in the country across multiple domains such as:

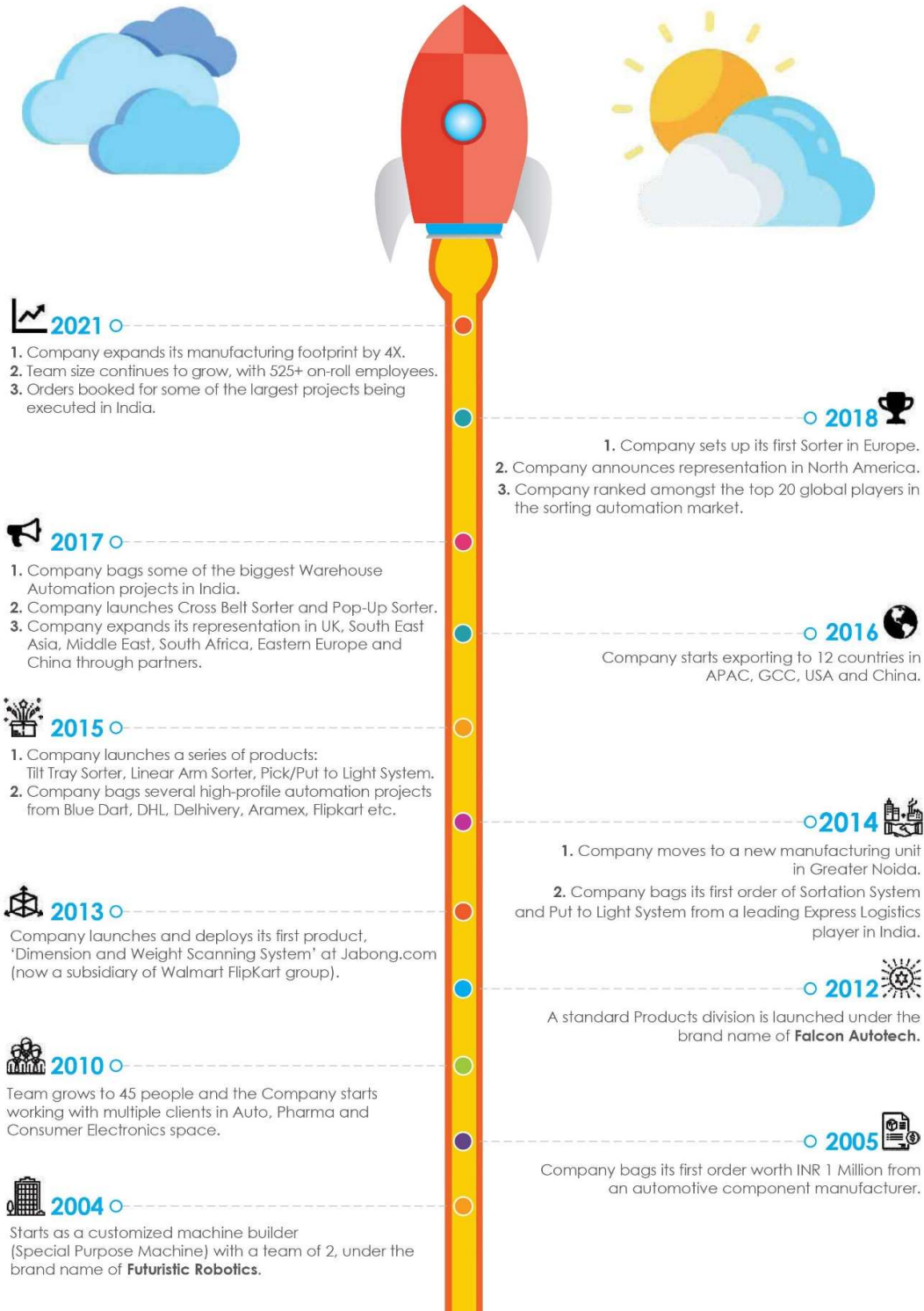
- ❖ Supply Chain Solution Designing and Consultation
- ❖ Hardware Design
- ❖ Electrical Design
- ❖ Electronics Design
- ❖ Software Development
- ❖ Manufacturing

Giving us an unprecedented ability to harness our true potential generated by culmination of Supply Chain Know-How, quality Hardware and flexible Software under one roof.





COMPANY HISTORY





KEY STATISTICS



04

Product Lines



525+

On-Roll Employees



290+

Engineering Team



400000+ Sq Ft

Manufacturing Space



20+

Tier A Clients



15+

Countries with Live Installations



1800+

Total Installed Systems Globally



35

Representations in 4 Continents and 35 Countries



1 Million+

Sorts Per Hour of combined Sorting Capacity
installed Globally



OUR PRODUCT PORTFOLIO

01

SORTATION SOLUTIONS



Cross Belt Sorter



Tilt Tray Sorter



Arm Sorter



Swivel Wheel Sorter



Sweep Sorter



Pusher Sorter



Pop Up Sorter

02

PICK/PUT TO LIGHT SYSTEMS



Packet Sortation



Multi-Line Order Consolidation



Order Picking



Store Order Fulfillment



Spare Parts Distribution



Kitting



Multi Order Pick Cart

03

DIMENSION & WEIGHT SCANNING SYSTEMS

STATIC DIMENSION & WEIGHT SCANNING SYSTEMS



Cubizon-R



Cubizon-R THRU



Cubizon-R Cross



Cubizon-R ECO

DYNAMIC CONVEYOR BASED SYSTEMS

OUR 8 MODELS





OUR PRODUCT PORTFOLIO

04

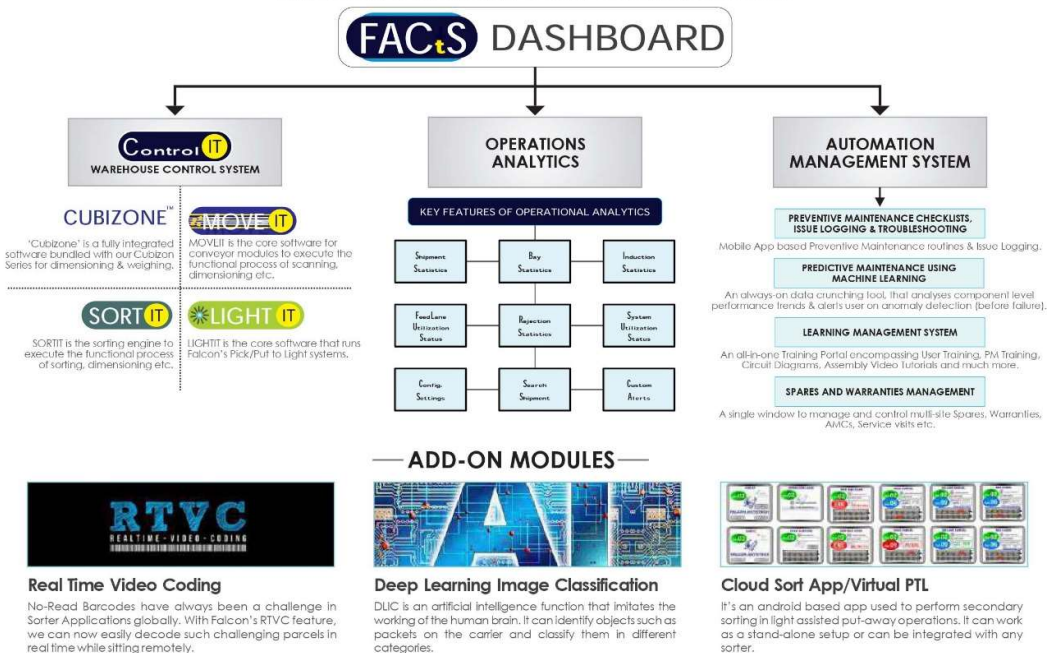
CONVEYOR AUTOMATION SOLUTIONS

BELT CONVEYORS	ROLLER CONVEYORS	SPECIAL APPLICATION SOLUTIONS
❖ PVC BELT ◆ Straight Conveyor ◆ Spacing Conveyor ◆ Accumulation Conveyor ◆ Strip Belt Merge- 30/45/60 ◆ Solid Belt Merge- 45 ◆ Incline Belt Conveyor ❖ MODULAR BELT ◆ Straight Conveyor ◆ 90 Degree Belt Curve ◆ 180 Degree Belt Curve 	◆ Straight Conveyor ◆ Gravity Roller Conveyor ◆ MDR Accumulation ◆ Angled Roller Conveyor ◆ Curved Conveyor ◆ Spur Merge Conveyor ◆ Swivel Wheel Diverter ◆ Pop Up Belt Diverter ◆ Pusher Unit 	◆ Aligning Conveyor ◆ One to Two Split Conveyor ◆ Two to One Merge Module ◆ Bulk Distribution-ARB ◆ Bulk Distribution-Telescopic ◆ Bulk Distribution-Electric Arm ◆ Telescopic Belt Conveyor 

05

SOFTWARE SUITE

FALCON'S AUTOMATION CONTROL SUITE





INDUSTRY SOLUTIONS

As a leading player in the intra-logistics automation space, Falcon Autotech continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in focused Industries and deliver world class projects in return.

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ❖ Piece Picking and Handling
- ❖ Case Picking and Handling
- ❖ Parcel Handling

FOCUS INDUSTRIES



E-COMMERCE RETAIL



COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



SPARE PARTS DISTRIBUTION

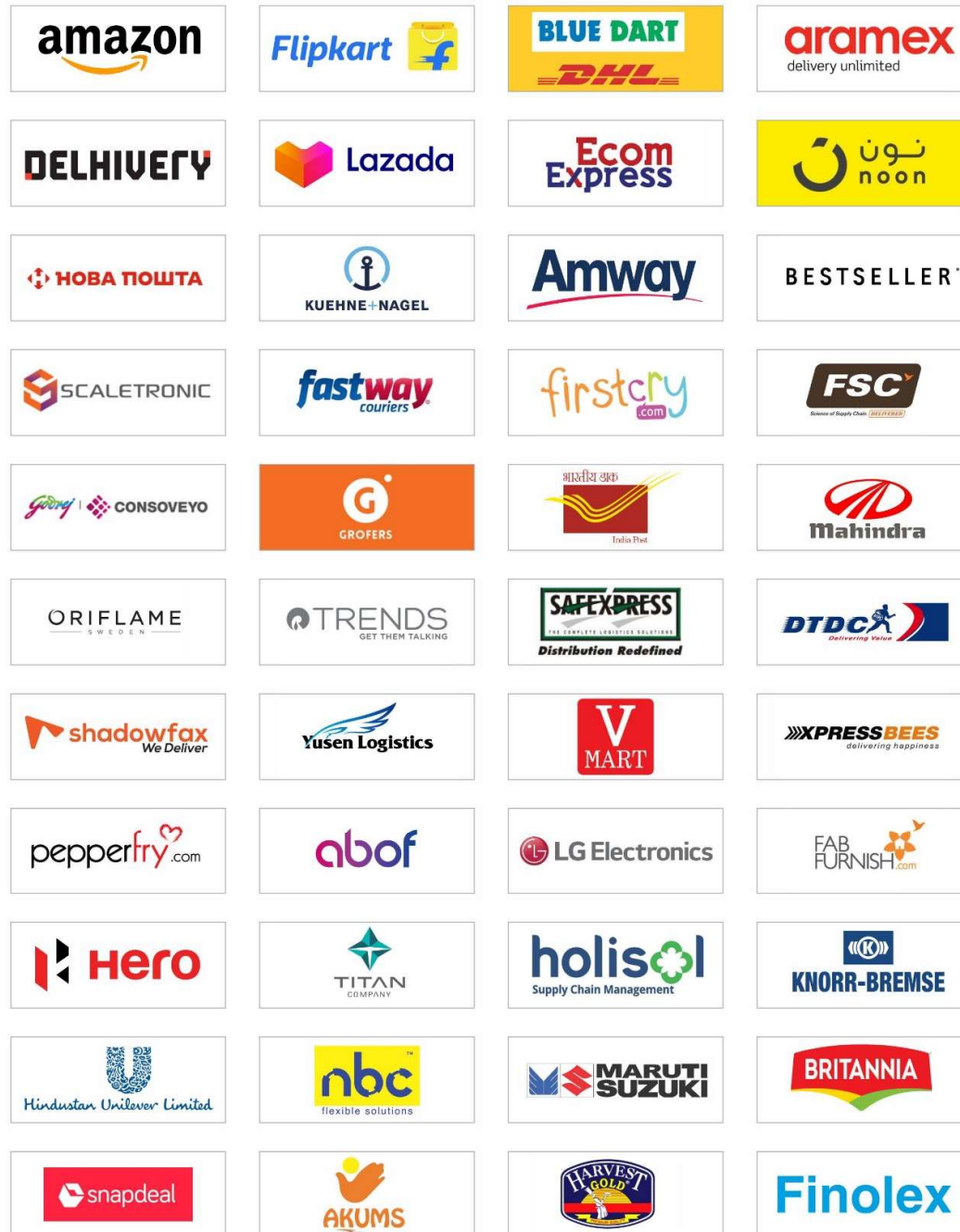


GROCERY DISTRIBUTION



SOME OF OUR GLOBAL CLIENTS

We believe in delivering high quality and standard worldwide. Here is a gist of our esteemed clients across various industry segments.





INDUSTRY SOLUTIONS

As a leading player in Intra Logistics Automation Space, Falcon Autotech continuously strives to improve the operational efficiency and accuracy for its clients through its domain knowledge consultancy and experience in addition to its wide range of Products and solutions.

In order to be able to live up to the high expectations set forth by our clients, the team at Falcon Autotech realizes the importance of taking up selective applications in Select Industries that lie in the Forte of Falcon Autotech and deliver world class projects in return.

FOCUS INDUSTRIES



E-COMMERCE RETAIL



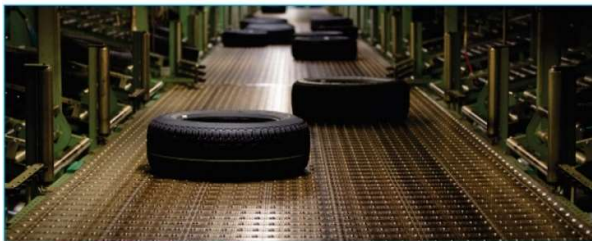
COURIER EXPRESS POSTAL



FASHION DISTRIBUTION



FFP (FOOD, FMCG, PHARMA) DISTRIBUTION



SPARE PARTS DISTRIBUTION



GROCERY DISTRIBUTION

FOCUS APPLICATIONS FOR FALCON AUTOTECH

- ◆ Piece Picking and Handling
- ◆ Case Picking and Handling
- ◆ Parcel Handling

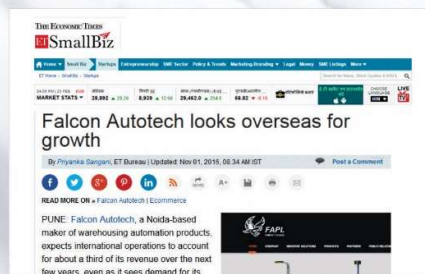


MEDIA COVERAGE

NO.1 LOGISTICS AUTOMATION COMPANY IN INDIA*



Falcon Autotech has achieved high recognition in the Intra-Logistics Automation world, "Global Logistics Automation Market" report published by Data Bridge Market Research has featured Falcon Autotech among top 20 in global list of Sortation Systems manufacturers. CNBC Awaaz, Awaaz Entrepreneur featured Falcon Autotech as fastest growing Logistics Automation company in India. Besides being featured in many important Daily Newspapers, we have also won the "Best Warehouse Automation Company" of the Year award at Express Logistics Supply Chain Conclave held in Mumbai in the Month of September 2015.



*Source: Report by Data Bridge Market Research



CLIENT TESTIMONIALS

“ Working with Falcon Autotech has been a very delightful experience for us, BLUE DART being a leader in the express logistics business, we expect the same kind of responsiveness, reliability, commitment & passion from our partners and Falcon has always delivered to their promise on all the solutions such as the Auto Sorter, Weight Dimension Labelling machine, Parcel Locker etc. All the technology interventions have been business critical for us.



Anil Khanna, Managing Director
DHL BLUE DART

“ Falcon is an organisation that has shown great agility and has helped us improve our service capabilities. Their sortation systems have helped us deal effectively with our growing volumes. We have found the Falcon team quite responsive to feedback and quick to incorporate changes into their systems that have helped build better, customized solutions for Delhivery.



Sahil Barua, CEO
DELHIVERY

“ Falcon Autotech has been efficient and proactive, particularly in meeting deadlines and their technical advancements. Falcon Autotech strike just the right balance between being a flexible, easy to deal with firm yet having all the key strengths and attributes. Every member of the team is honest, reliable, responsive, personable and very knowledgeable. I wish them good luck for all the future endeavours.



Sanjeev Saxena, Director
ECOM EXPRESS

“ I believe you guys earned it with your hard work and dedication. I believe the products you have designed demonstrate innovation and lateral thinking. Also the management has a flexible and the customer centric approach. It has been a delight to work with you and the team.



Amitava Saha
CEO, firstcry.com



3. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers** globally. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolio in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **6 Million Products per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.

Born2Invest
Save It. Invest It. Prosper.

Incas-SSI Schaefer merger creates leading player in Italy's intralogistics market

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Moreover, the transportation management segment of the logistics automation industry is expected to have a higher CAGR. This is due to factors associated with the growing need to deliver goods in a timely manner and to reduce transportation costs.

Moreover, automotive will be the leading segment of the industry from 2018 to 2023, with an increasing demand for automated production systems and flexible logistics systems used in manufacturing and supply of vehicle fleets.

The MarkestandMarkets research report has identified SSI Schaefer as one of the key players in the logistics automation market worldwide during the forecast period, along with Dematic Corporation (US), Murata Machinery, Ltd. (Japan), Jungheinrich AG (Germany), and Falcon Autotech (India), among others.

This year's largest market for logistics automation is estimated to be North America, according to the research report. Regardless, Incas and SSI Schaefer's merger could help them compete in their main market.

(Featured image by [DepositPhotos](#))

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Not secure | worldindustryinsights.com/30269/global-parcel-sortation-system-market-analysis-size-share-growth-trends-and-forecast-2019-2026-siemens-vanderlande-beumer-group

Global Parcel Sortation System Market Analysis, Size, Share, Growth, Trends, and Forecast 2019-2026: Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives

BY ANN.CASTRO ON AUGUST 12, 2019

The global "[Parcel Sortation System market](#)" research report is a pervasive research report that touches the most vital parts of the Parcel Sortation System market that is necessary to be grasped by a professional or even a layman. The market research report enlightens one regarding few of the important aspects such as an overview of the Parcel Sortation System product, the growth factors enhancing or hampering its development, application in the various fields, major dominating companies, genuine facts, economic situation, and geographical analysis. The research report endows the data concerning the dynamics that propel the growth as well as the supply and demand chain of the product on a global basis.

Click here to access the report: www.reportsbuzz.com/request-for-sample.html?repid=50822

The international and global stand of the Parcel Sortation System market is also briefly mentioned in the research report based on the performed statistical and thorough market analysis. The information mentioned in the research report gives a qualitative and quantitative view of the overall market. The statistical analysis of the market helps analyze the supply, demand, production, maintenance, and storage costs of the product. The data regarding some of the prevalent players Siemens, Vanderlande, Beumer Group, Intelligrated, Bastian Solutions, Fives, Dematic, Interroll, Muratec, Invata Intralogistics, Viastore, Okura Yusoki, GBI Intralogistics, Equinox Mhe, Pitney Bowes, Falcon Autotech, OCM SRL, Solystic, Bowe Systec, Intralox are also detailed in the current case study.



4. Reference projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Piece Picking and Handling, Case Picking and Handling & Parcel Handling
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

4.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 parcels per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before parcels enter the sortation system. The parcels are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –





4.2 Project 2- (CEP Client, Riyadh)

In 2019, customer selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing of parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

Site Picture –





4.3 Project 6- (Client – Courier Express Parcel, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 parcels per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –





4.4 Project 4- (CEP Client, Sydney)

Solution is designed for handling a throughput of 16,000 parcels per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the parcels at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. System is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight parcels.

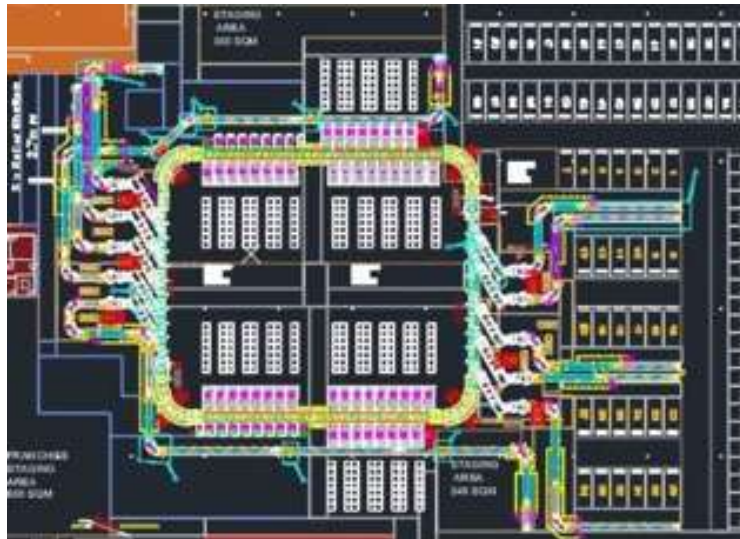
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize parcel.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –





4.5 Project 3- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

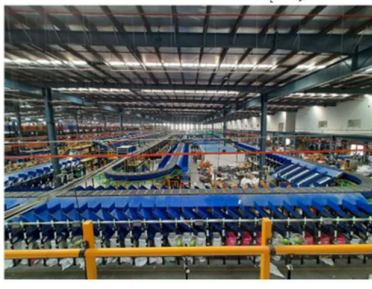
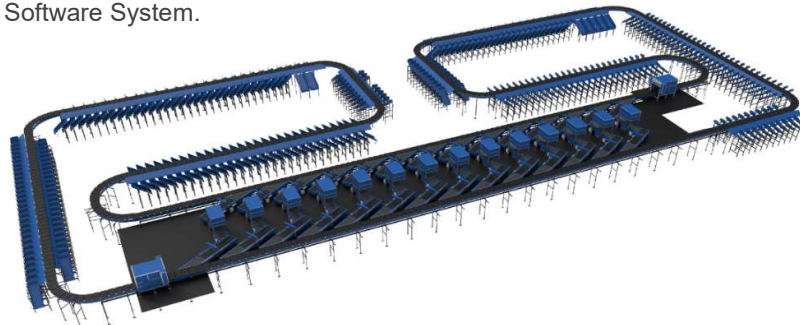
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Presort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system
- WCS Software System.





4.6 Project 5- (Ecommerce Client, India)

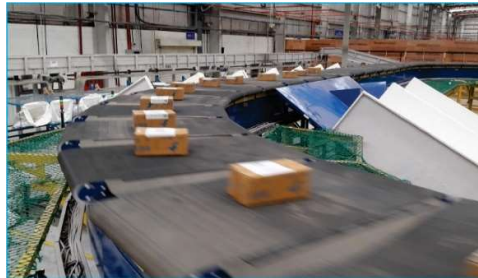
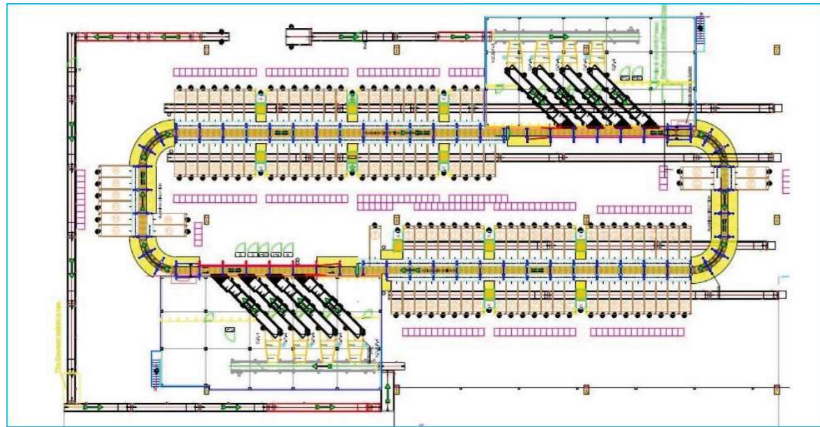
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



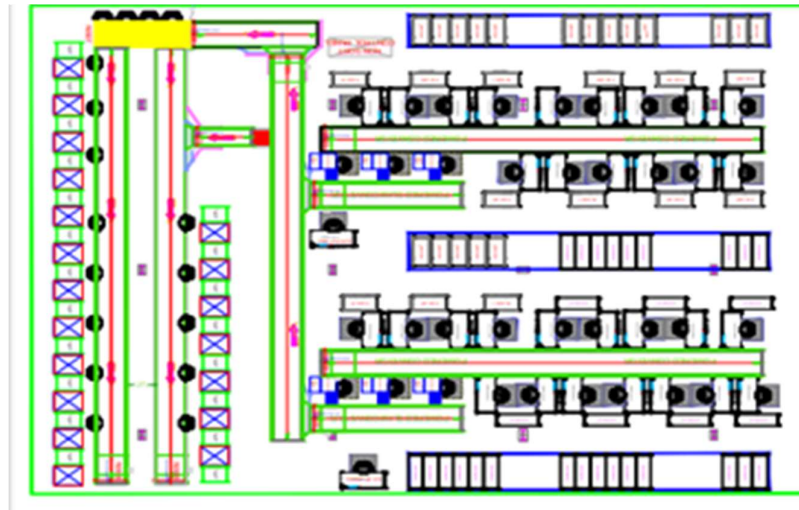
Layout & Pictures_ Project 1



4.7 Project 7- (E-commerce, India)

Solution is designed to handle the parcels Inbound and Outbound for packaging and sorting. Types of conveyors have used in this solution are as follows :

- Belt Conveyor
- Roller Conveyor
- Goose Neck Inclined





5. Handled Parcel Spectrum

5.1 Product size Conveyable on the entire system.

Max Length	mm	600
Max Width	mm	600
Max Height	mm	600
Max Weight	Kg	18
Min length	mm	150
Min Width	mm	150
Min Height	mm	150
Min Weight	gm	100

5.2 Products to be loaded on Conveyor/Sorter shall have the following characteristics:

1. Centre of Gravity of items must not move during conveyance or sorting.
2. Item must not have magnetic content; otherwise behaviour of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Products shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic conveying/sorting system or damaging the system. They must be robust enough to avoid disintegration of container material and loose of contents in the conveying/sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The products must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)



5.3 Parcel not loadable on the system

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.
3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile products with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

5.4 Types of Products

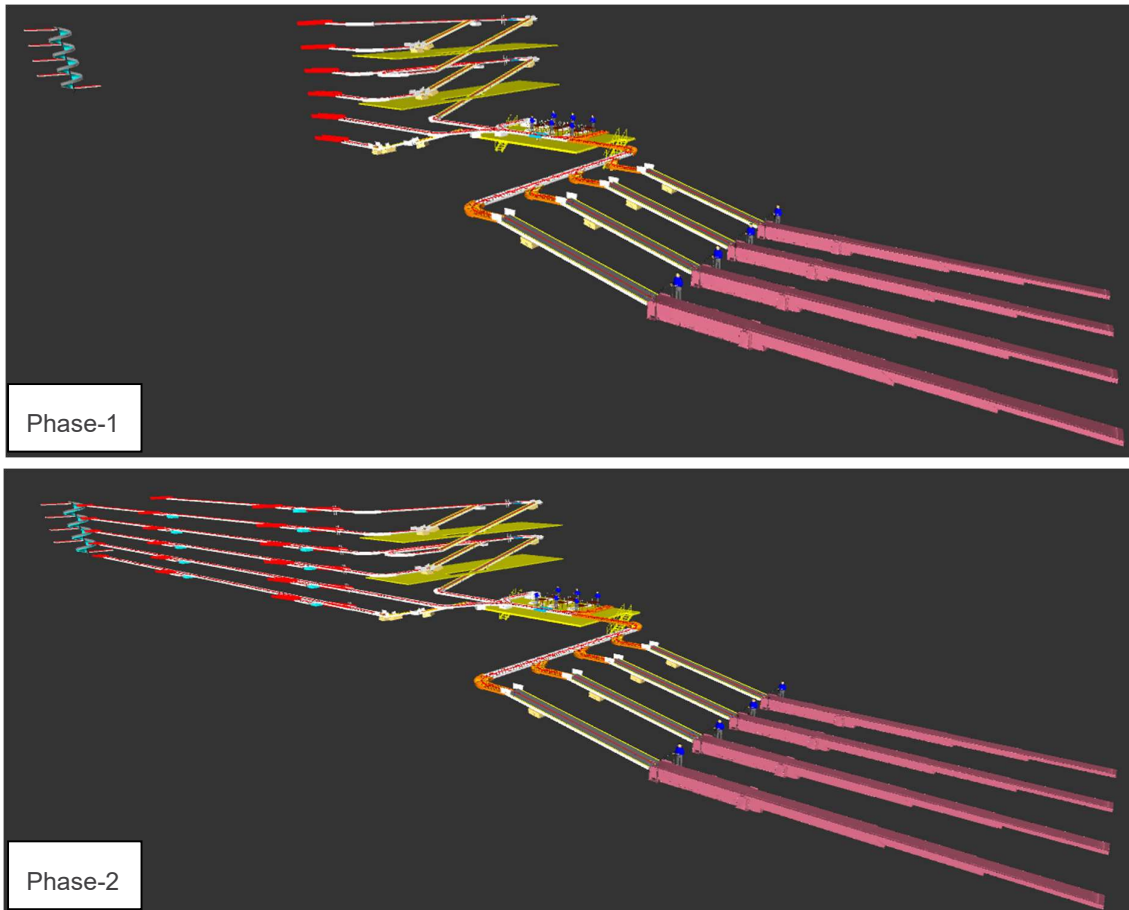
Cartons	Two Types of Cartons Min: L-600 X W-600 X H-600 MM Max: L-150 X W-150 X H-150 MM



6. Proposed System Description

6.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing and acceptance testing of the complete automation system to handle Cartons as per specifications shared by Landmark.



6.2 Summary of the System

1. **Infeed:** Cartons will be manually loaded on four telescopic belt conveyors, Physical barrier are present to restrict oversize products, further, parcel is transferred to inclined belt conveyor that transport the products primary scanning and distribution zone.
2. **Primary Scanning & parcel segregation:** Cartons are now scanned through side barcode scanner & get diverted to either levels, Parcel those contain different SKUs inside them & products those miss scanning will be diverted to manually processing zone.
3. **Manual processing Area:** Cartons those are diverted to this area, are manually opened & different SKUs are manually loaded on powered roller conveyor, Now these products are scanned & diverted to respective levels.



4. **Floor Wise Product segregation:** Products are transported & diverted via Set of conveyors & SWEDI diverters. These powered conveyors transport products to desired levels as per logic decided by Landmark.

Phase 1 - Products are dropped at the starting point of mezzanine at respective levels.

Phase 2 – Products as transported through the length of the particular level & also diverted to respective Aisle through Pop up diverters.

5. **Outbound:** After the manual picking is complete, Cartons are carried to the other end of the mezzanine & loaded on Downstream Powered belt Spiral conveyor.

6.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

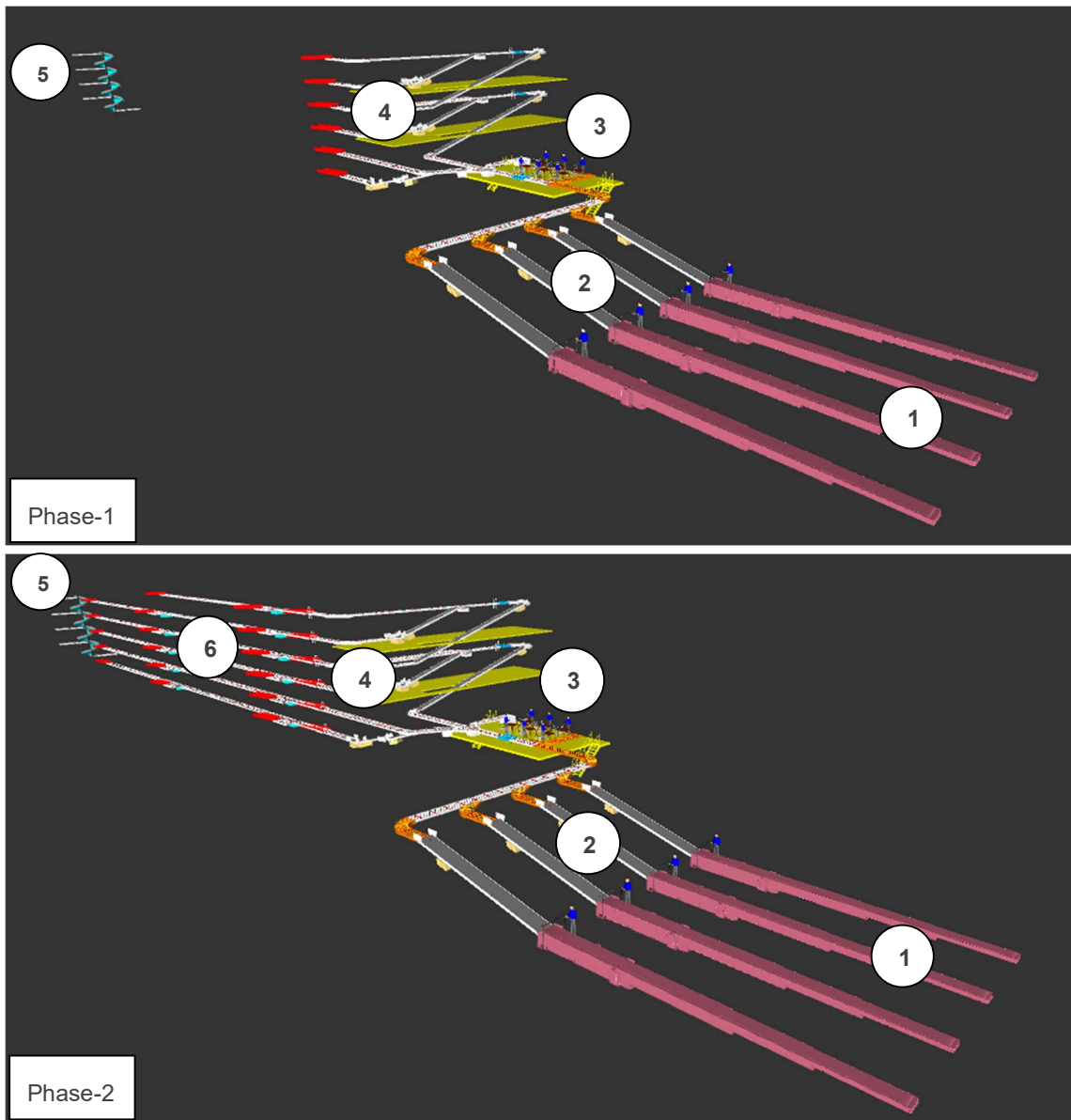
1. High operational throughput
2. Maximized sorting capacity throughout at dispatch sorter to meet the requirement.
3. Low occupancy of floor space in the building
4. FALCON's system can adapt with changing business requirements by adjusting its speed to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear

7. Proposed System Capacity

Activity	Cartons per hour
Conveyor Automation	1200
Powered belt spiral conveyor	1000



8. Layout View



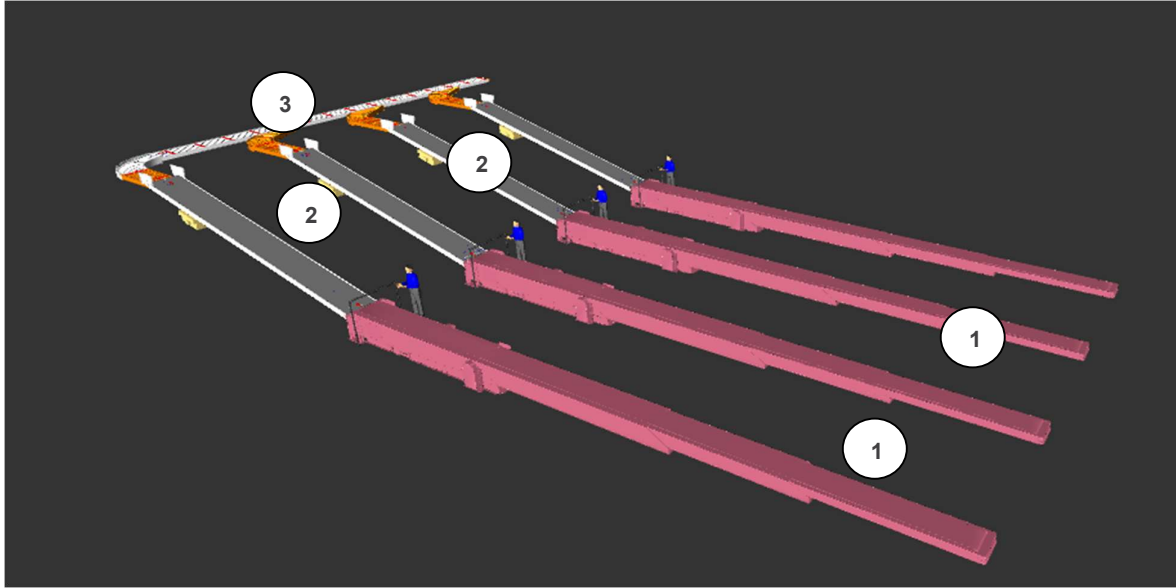
Legend

- 1 – Telescopic Belt Conveyors
- 2 – Inclined belt conveyor
- 3 – Primary scanning zone & manual processing zone
- 4 – Conveyor Automation for Inbound distribution
- 5 – Downstream Powered Spiral Conveyor
- 6 – Conveyor Automation for Aisle Distribution



9. System description

9.1 Infeed System

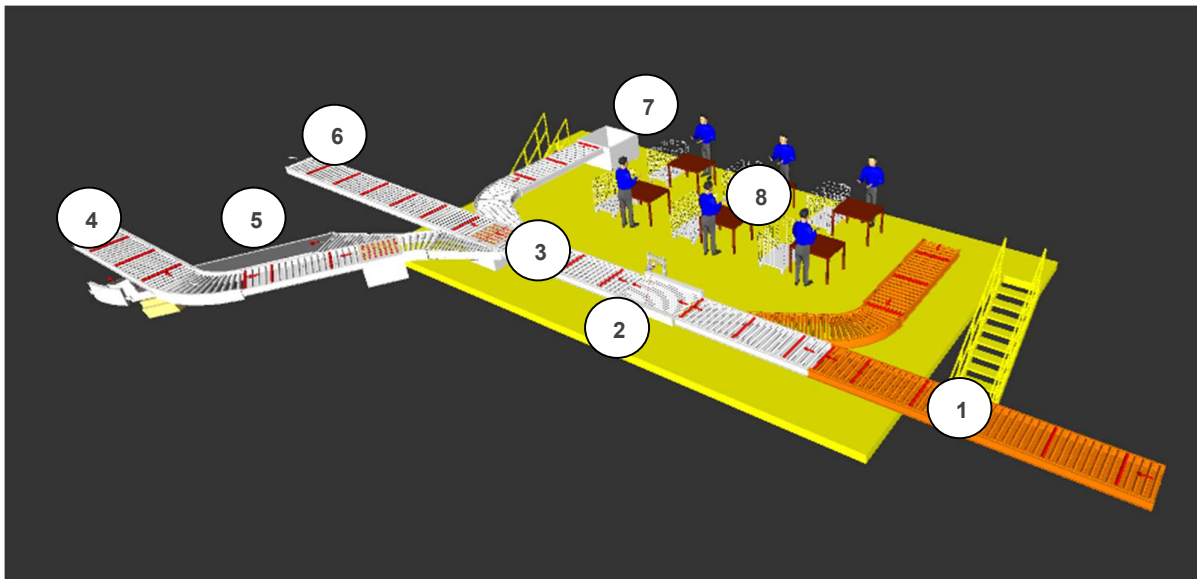


Legend

- 1 – Telescopic belt conveyor
- 2 – Inclined belt conveyor
- 3 – Powered roller conveyor



9.2 Scanning & manual processing zone

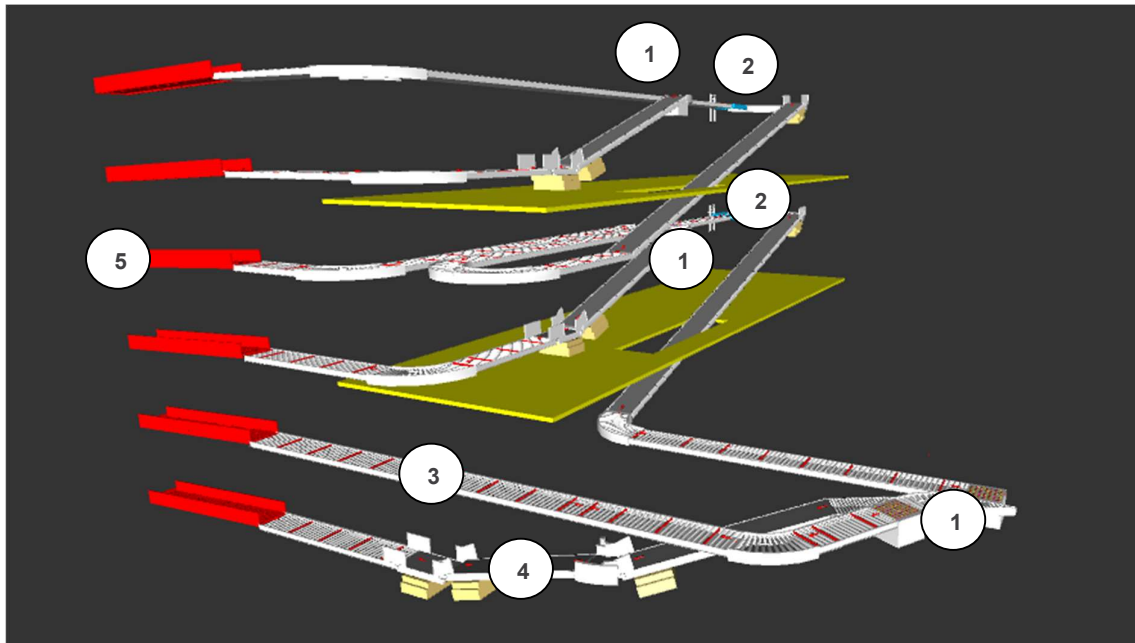


Legend

- 1 – Powered Roller Conveyor
- 2 – Side Barcode Scanner
- 3 – SWEDI Diverts
- 4 – Carton diversion @ L+1
- 5 – Carton diversion @ L
- 6 – Carton diversion @ L+2 to L+5
- 7 – Carton diversion @ manual processing area
- 8 – Manual processing area



9.3 Carton distribution system

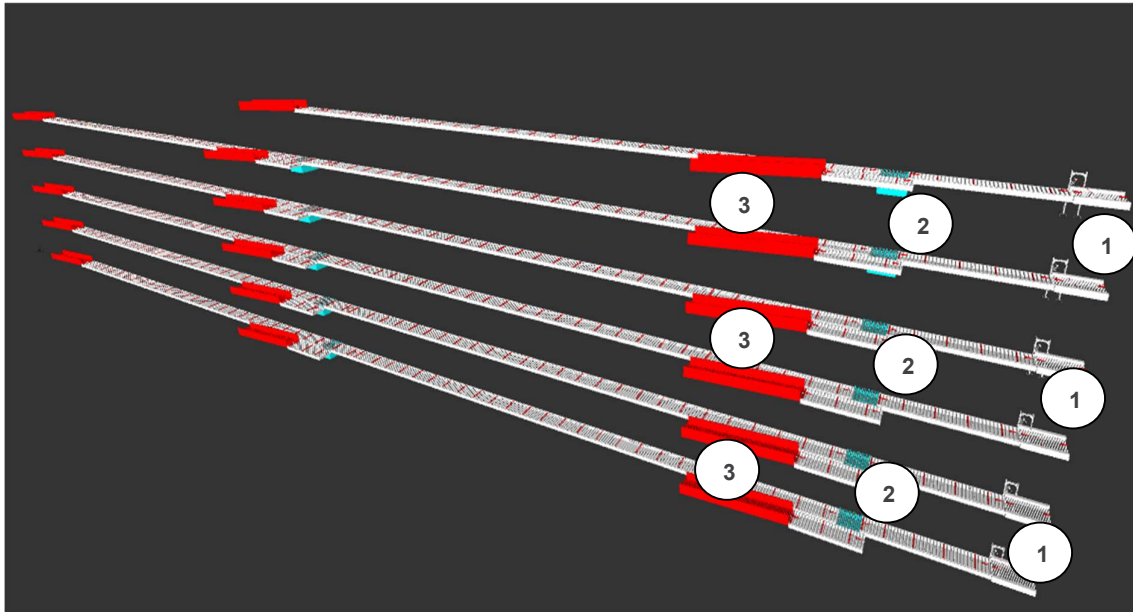


Legend

- 1 – SWEDI diverter
- 2 – Side barcode scanner
- 3 – Powered roller conveyor
- 4 – Powered belt conveyor
- 5 – Gravity roller conveyor (all levels)



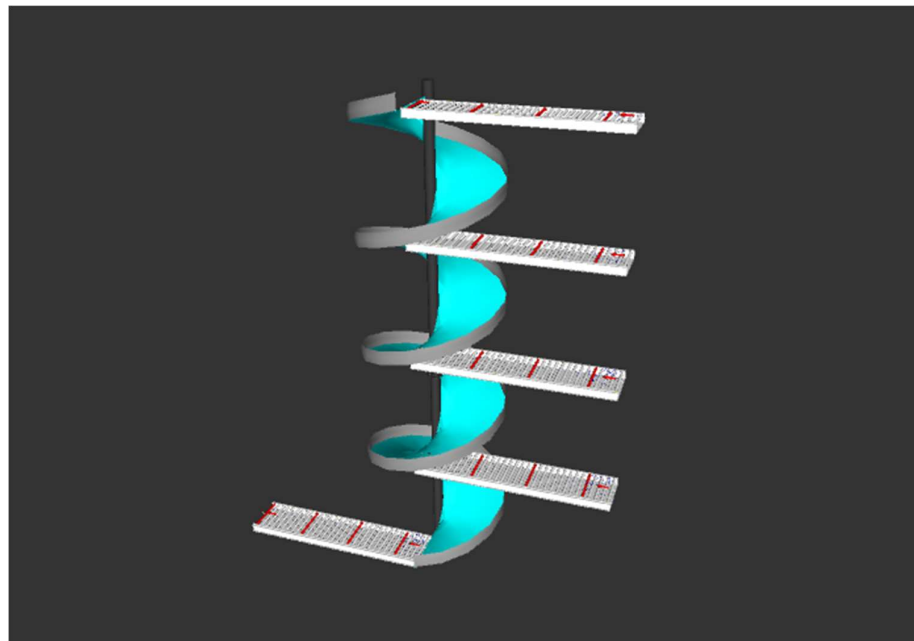
9.4 Carton Distribution into Aisle (Phase-2)



Legend

- 1 – Side barcode scanner
- 2 – Pop up sorters
- 3 – Gravity roller conveyor

9.5 Outbound movement



Legend

- 1 – Downstream powered spiral conveyor



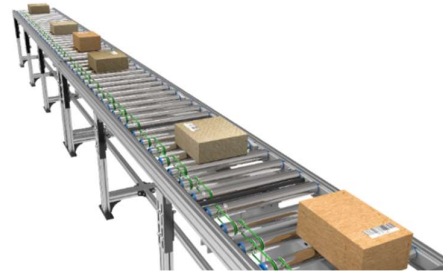
10. System Components

10.1 Powered Roller Conveyor

Falcon's Roller conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	800/600
Conveyor Height	mm	Variable as per design shared
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Poly V Belt Drive
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Load capacity/unit length of Conveyor	kg/m	50
Roller Span	mm	Variable
Roller Diameter	mm	50
Pitch of Rollers	mm	75/100
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions



10.2 Idler Roller Conveyor

Falcon's Roller conveyors require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's idler roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	800/600
Conveyor Height	mm	As per Layout requirement
Load capacity per unit length of Conveyor	kg/m	50
Roller Span	mm	approx. 600
Roller Diameter	mm	approx. 50
Pitch of Rollers	mm	75
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions
Type of Guides	Type	Yes
End Stopper	yes/no	Yes



10.3 PVC Belt Conveyor

Falcon's Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

Some Salient Features of Falcon's Belt conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
MOC of Belt	Type	PVC
Belt Thickness	mm	3
Belt Finish	Type	Matt Finish
Belt width	mm	800/600
Belt Joint Type	Type	Vulcanized, Endless Belts
Belt Make/Model No.	Make	FORBO/DERCO
Chassis	mm	MS, 3mm Thick, Powder Coated-75 Microns
Idler Roller Pitch on running side (Top)	mm	480
Idler Roller Pitch on return side (Bottom)	mm	1200
Idler Roller Specs	Spec	Dia.50.8, shaft-12 mm, bearing-6301
End Rollers	Spec	Dia 81.2 x 2.85 Thick, 30mm Shaft Dia, & Bearing Size-UCFH206D1, MOC-EN9, Zinc Plated -25 microns
Drive Rollers	Spec	OUTER Dia 178 PIPE Dia 168x5 Thick, 40mm Shaft Dia, & Bearing Size-UCF208D1, MOC(SHAFT)-EN9, NEOPRENE RUBBER COATING
Tensioning Rollers	Spec	Dia 81.2 x 2.85 Thick, 25mm Shaft Dia, & Bearing Size-UCT205D1, MOC-MS, Zinc Plated -25 microns



Take Up Rollers	Spec	Dia 81.2 x 2.85 Thick, 25mm Shaft Dia, & Bearing Size-6305-LLU, MOC-MS, Zinc Plated -25 microns
Type of Drive	Type	Rubberized
Angle of wrap/Contact - Belt with Drive pulley	Deg	>180 deg
Motor Shaft	MOC	EN9 Shaft material,
Conveyor Under guarding	Spec	MOC-MS, 0.8 to 1mm thickness, Steel Stiffeners at every 600mm to avoid the sagging due to self-weight
Load capacity per unit length of Conveyor	kg/m	50
Type of Guides	Type	Steel guides at 50mm/400mm height basis the position of conveyor on both sides from Conveyor Belt Surface.
Drive Power Rating	kW	Motor Rating depends on Length of the Conveyor
Type of Motor	Type	AC Geared Motor
Ingress Protection	Type	IP 55
Type of Drive System	Type	Direct Shaft Mounted Motor/ Flange Mounted
Type of Drive	Type	Chain drive/Torque Arm type
Gear Motor	make	Sew/Nord
Drive	make	Lenze/Siemens/Omron/Allen Bradley
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions

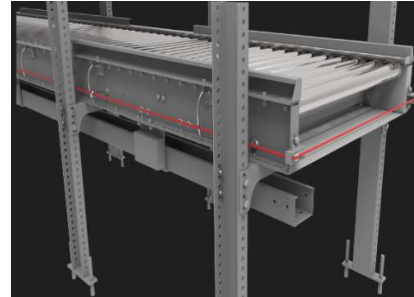


10.4 Accumulation Conveyor

Falcon's Accumulation conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's accumulation conveyor:

- Low Noise
- Maximum uptime.
- High safety standards.
- Fastest ROI
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	Mm	800/600
Conveyor Height	Mm	As per Layout requirement
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Round/Poly V Belt Drive
Insulation Class	Type	Yes
Conveyor Speed	m/s	Variable speed to meet TPH requirements
Load capacity per unit length of Conveyor	kg/m	50
Roller Span	Mm	Approx. 600
Roller Diameter	Mm	Approx. 50
Pitch of Rollers	Mm	75
Zone Length	Mm	825
Type of Mounts	Type	Fixed Heights Legs with Grouting Provisions
Type of Guides	Type	Steel guides at 50mm height basis the position of conveyor on both sides from Conveyor Belt Surface.



10.5 Powered Roller Turns & Merges



10.6 ZTR Curves (Pic for ref.) & Roller Aligning Conveyors



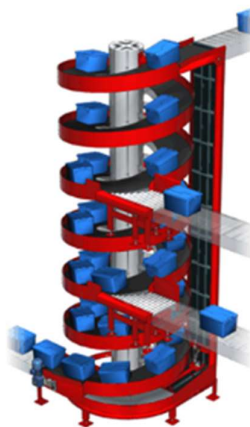


10.7 Telescopic Belt Conveyor



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Suitable Carton Size	MM	Min: 150 x 150 x 150 Max: 600 x 600 x 600
Suitable Carton Weight	Kg	Min 5, Max 25
Base Length	Mm	6000
Extended Length (Total Length)	Mm	12000
Conveyor Belt Running Speed	m/min	18-30
Motor Rating	Type	0.75 Kw, 415V AC, 50 Hz
Load Capacity	Kg/m	50
Communication	Mm	2 x RS243, 1 x USB2.0, Ethernet

10.8 Powered Spiral conveyor





10.9 Pop-Up Sorting Unit

Belt Popup device is used in system for 90° transfer of products.

Some Silent Features of Pop-Up:

- Low Noise
- Maximum uptime.
- Electric based popup.
- Minimal maintenance
- High safety standards.



Specification	UOM	Remark
Manufacturer Name	Name	Pulse
Type – Equipment	Type	Stand Alone
Type – Operation	Type	Yes
Mechanism for Pop-Up	Type	Electric
Type of Motor	Type	DC
Ingress Protection type	Type	IP 54
Provision for Carton Box / Carton Detection Sensors	Yes/No	Yes
Load Capacity of Carton Box	Kg	18
Sorting Capacity	PPH	1200



10.10 Swivel Wheel Sorting Unit - Pulse

Swivel Wheel divert is used in the system for the smooth transfer of products.



Specification	UOM	Remark
Manufacturer Name	Name	Kyowa
Dimensions	mm	As per design
Type - Equipment	Type	Merge/Sorting
Type - Operation	Type	Sorting
Mechanism for Swivel	Type	Electric servo motor
Roller drive	kW	Powered smart wheel
Roller pitch (L x W)	Type	110 x 120
No. of divert sections	Yes/No	2
Max. product weight (kg/m2)	kg/m2	25
Throughput	PPH	1200
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions



10.11 Side Barcode Scanners

The system is integrated with automated inline side barcode scanning system capable for scanning 1D barcodes with high accuracy.



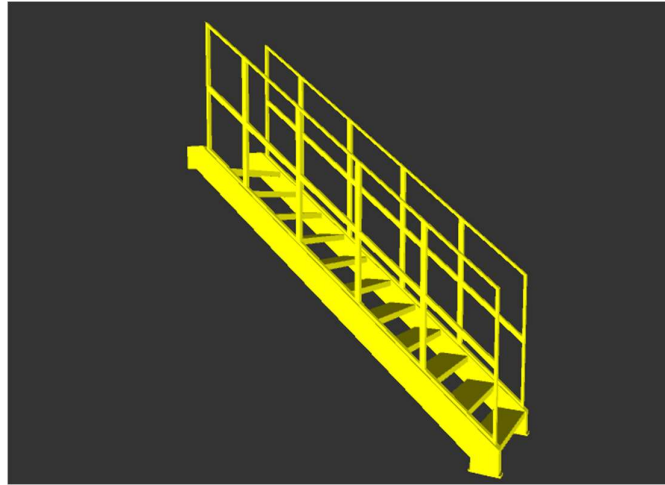
Specification	UOM	Remark
Manufacturer Name	Name	Cognex/Sick
Barcode Type	Type	1D
Module Size	Mils	12
Carton Sizes	Mm	Max: 600 x 600 x 600 Min: 150 x 150 x 150
Number of Codes	Nos	1
Location	Side	Left
Orientation	Type	Omnidirectional
Color of bars	Color	Black
Under Foil	yes/no	No
Code Length	Mm	50
Code Height	Mm	15

*Note Barcode position assumed to be fixed



10.12 Ladders

Total 2 Nos. Ladders in the system



10.13 Operator's Platform

Operator platform is planned for Manual processing area & primary scanning area.





11. Proposed System Technical Details (Phase-1)

11.1 Mechanical equipment

S.no.	Qty.	Description	Value
1	1	Falcon Conveyor Automation 04 X Telescopic belt conveyor 04 X Physical barriers 03 X Side barcode scanner 26 X Powered roller conveyor ~ 185 m 13 X Powered belt conveyor ~ 122 m 06 X Gravity roller conveyor ~ 38 m 06 X Accumulation Conveyor ~ 23 m 03 X Aligning conveyor ~ 4 m 04 X SWEDI sorter 05 X Merge 13 X 90° Turn 01 X 60° Turn 05 X 30° Turn 05 X Y Merge 01 X Powered Spiral conveyors	~ 800 mm ~ 800 mm from Mezz Top
2	1	Steelworks Staircase Mezzanine ~ 104 Sqm Platform ~ 158 Sqm	02 Nos. 01 Nos. 02 Nos.
3	1	Standard Falcon's Safety Equipment	Included

11.2 Electrical Equipment

Electricals		
1	Consists of Main Power Distribution panel Main Control Panel Network Switches Field Cabling	Included

11.3 Control System

Components		
1	Consists of Siemen's PLC based Control System Industrial Switch	1 Nos. As per requirement



12. Proposed System Technical Details (Phase-2)

12.1 Mechanical equipment

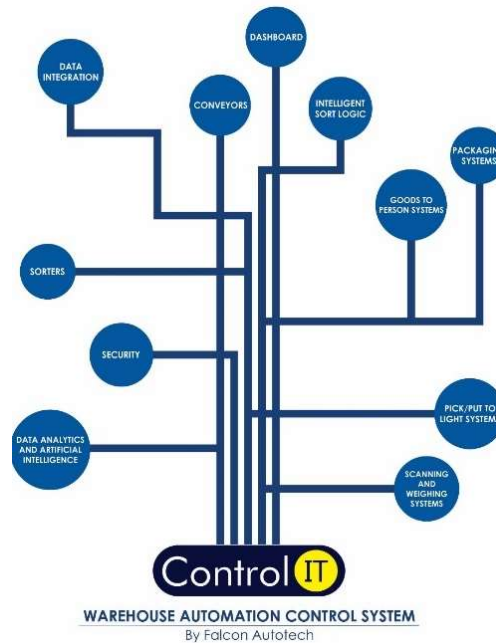
S.no.	Qty.	Description	Value
1	1	Falcon Conveyor Automation 06 X Side barcode scanner 34 X Powered roller conveyor ~ 395 m 17 X Gravity roller conveyor ~ 85 m 06 X Accumulation Conveyor ~ 18 m 06 X Aligning conveyor ~ 9 m 11 X Pop-Up diverter	~ 800 mm ~ 800 mm from Mezz Top

12.2 Electrical Equipment

Electricals		
1	Consists of Control Panel Field Cabling	Included



13. Falcon's WCS CONTROLIT



Key Values

Supports Multiple Use Cases

- Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more

Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions

Supports various peripheral Equipment Inputs and Outputs

- Automated Scanners, Manifest Printers, Label Applicators and many more

Ability to define Custom Business Logics

- Cut Off Timings, Quantity/Weight and Volume Thresholds etc

Security

- User profiling with Role Right Mapping
- Database Encryption
- Secured and Protected Communication between WMS and WCS layers

Supported Protocols

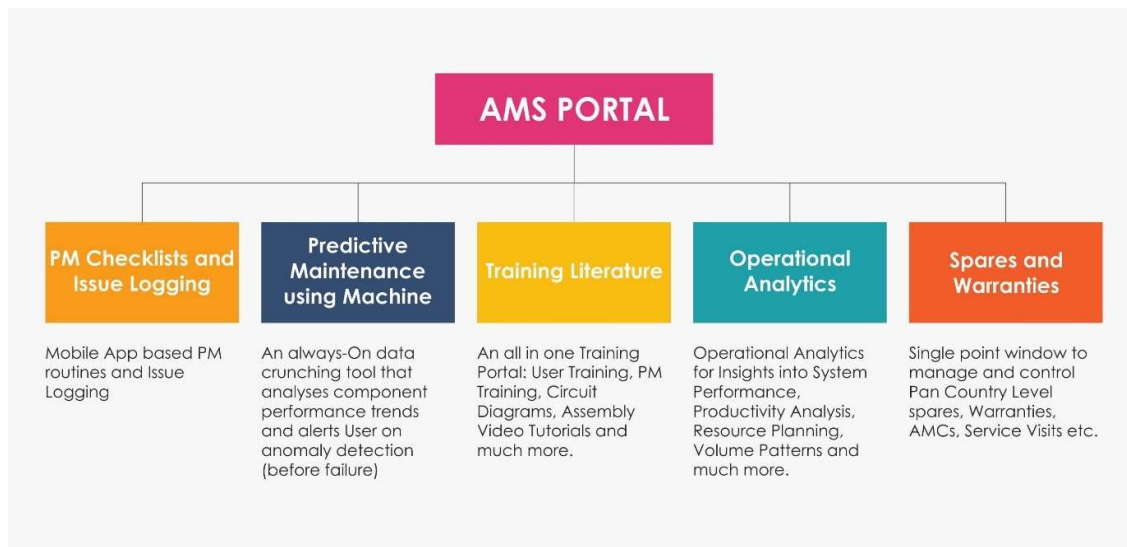
API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol



14. Key Component Makes

Component	Make
PLC & RIO	Siemens
VFD	Siemens/Omron/Lenze/AB
Tower lamp	Leuze
Switchgear	Schneider
Geared Motors	SEW
MDR	Pulse/ Itoh Denki
Scanner	Sick/ Cognex
Diverter	Falcon/Kyowa

15. Automation Management System (AMS)





15.1 Issue Logging

The first screenshot shows the 'Issue Logging(T-214)' screen with the 'ADD ISSUE' tab selected. It contains a form with the following fields: Problem Observed, Reason, Sub Reason, Sub Sub Reason, Corrective Action, Time Taken to Resolve, Onsite Customer Engineer, and Onsite FAPL Engineer. Each field has a '+' icon to its right. The second screenshot shows the 'APPROVE ISSUE' tab with a list of issues. The third screenshot shows a modal dialog box for a specific issue.

Issue ID	Problem	Reason	Corrective Action	Date
1489	PTL Not Working	due to green light not blinking	programme done	29/12/2017 12:00:00 AM
109	sorter stopped			
107	Trolley Not Tilting			
106	Trolley Not Tilting			
96	Trolley Not Tilting			
95	wrong sorting from feed lane-1			

15.2 Preventive Maintenance Check List

The first screenshot shows the 'CL(T-214)' screen with the 'Selected Date : 2018-02-27'. It has tabs for HOURLY(4), DAILY(111), WEEKLY, and MONTHLY. The 'Operations' section is expanded, showing 'Collection Bin' with a status of 'OK'. The 'Induction' section shows 'Wide Conveyor' with a status of 'NOT OK'. The 'Rejection' section shows 'Last Collection Bin' with a status of 'NOT OK'. The 'PLC Logs' section shows 'PLC' with a status of 'OK'. The second screenshot shows a modal dialog box for a specific item.

Item	Sub Item	Observation	Remarks
Main Sorter	master & slave motor	Check for oil leakage mark in gear box	



15.5 Operational Analysis



16. Program Organisation

16.1 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Landmark
3. Scheduling of the resource management.
4. Management of the risks and opportunities.
5. Management of the list of anomalies or reservations.

16.2 Program Schedule

1. 18-20 weeks Ex-works Greater Noida



17. Commercial Terms

17.1 Price Sheet

Phase-1

S. No	Component	Set	Price (INR)
1	Swedi Sorter	1	₹ 18,67,040
2	Telescopic belt conveyor	1	₹ 54,00,000
3	Conveyors	1	₹ 3,19,85,238
4	Spiral Conveyors (OB)	1	₹ 95,73,606
5	Steelwork	1	₹ 56,10,000
6	Software Packages	1	₹ 6,00,000
7	Toolkit	1	₹ 36,000
8	Packaging & forwarding	1	₹ 8,97,966
9	Installation & Commissioning	1	₹ 22,44,914
10	Project Management and Engineering Charges	1	₹ 12,16,029
Total (INR)			₹ 5,94,30,791

Phase-2

S. No	Component	Set	Price (INR)
1	Pop-Up Diverters	1	₹ 32,26,667
2	Conveyor Automation	1	₹ 3,56,45,049
3	Packaging & forwarding	1	₹ 7,77,434
4	Installation & Commissioning	1	₹ 19,43,586
5	Project Management and Engineering Charges	1	₹ 10,39,818
Total (INR)			₹ 4,26,32,554

The Total Price is to be understood

- Ex Works – Greater Noida, India
- Freight charges shall be extra at actual
- GST will be charged extra @ 18%
- Including Packing and Forwarding
- Duty unpaid
- Installation and Commissioning: Mentioned above.
- Price is valid for 60 days from the date of proposal

Our Price does not include taxes, and duties, import duties, charges, fees, taxes etc. charged by authorities outside India for deliveries and services. Same shall be borne & paid by client itself.

*Material Unloading & Laydown area is in customer scope



17.2 Payment Terms

The Supplier Quotation is based on the following payment schedule:

Stage 1: 50% Advance along with PO.

Stage 2: 40% along with 100% Taxes prior to dispatch.

Stage 3: 10% After Installation and commissioning within 7 Days.

18. Warranty Period

Falcons offered System comes with a standard warranty of 1 year (Starts from the date of beneficiary use) and post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.



19. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- Servers
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms- If not specified
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier's scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.